ל1 — Identifying 3: The Native Simplicial Chart

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Abstract

Situation. Information geometry is usually introduced as an analyst's project: probability distributions live on a manifold with a normalization constraint, the natural metric (Fisher–Rao) is positive-definite but awkward to differentiate against, and the standard move — eliminating one coordinate to remove the constraint — silently breaks the permutation symmetry that the simplex obviously has. Most of the technical effort in introductory expositions is bookkeeping to repair what the elimination broke.

Task. Find a coordinate system in which the simplex's intrinsic structure is exposed rather than concealed.

Action. Refuse the elimination. Keep the redundant N-th coordinate, treat the constraint $\sum p_i = 1$ as a gauge fixing, and work on the overcomplete frame $\mathbb{R}^N$. Read probabilities as squared amplitudes: $p_i = a_i^2$ with $\sum a_i^2 = 1$. The amplitude vector a is a point on the positive orthant of the unit sphere; the probability simplex is its squared shadow.

Result. Fisher–Rao is the pullback of the round sphere metric through the square-root map — sphere geometry in probability coordinates. The tangent space at any interior point of the simplex is the zero-sum hyperplane H_0, a genuine Euclidean vector space. The Fisher–Rao isometry group at the uniform distribution is $\mathrm{SO}(H_0)$. The chart carries an intrinsic vector algebra — inner product, cross product, Rodrigues rotation — that reproduces ordinary three-dimensional Euclidean vector algebra exactly in the tetrahedral case $N = 4$. This paper presents the chart and walks it through worked examples for $N = 2, 3, 4$.

This is ל1 of the trilogy *Identifying 3*. ל2 develops the cumulant tower of the categorical log-partition on this chart; ל3 reads the chart through Pontryagin character orthogonality. ל1 supplies the toolkit they share.

Keywords: information geometry, statistical manifold, probability simplex, categorical distributions, Fisher–Rao metric, square-root coordinates, amplitude sphere, zero-sum hyperplane, Bhattacharyya coefficient.

Probability distributions over N outcomes live on a simplex — a triangle for $N = 3$, a tetrahedron for $N = 4$ — that sits awkwardly inside N-dimensional coordinates because the probabilities must sum to one. Take the square root of each probability and the awkwardness goes away: the simplex becomes the positive octant of a sphere, and the Fisher–Rao metric reduces to ordinary sphere geometry. No coordinate elimination, no Lagrange multipliers, full permutation symmetry preserved by construction. For four outcomes, the tangent space is three-dimensional and the chart's intrinsic algebra is literally ordinary 3D vector calculus — inner product, cross product, rotation — usable as-is.

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Notation

Symbol	Meaning
N	Number of categorical outcomes; integer ≥ 2.
Δ^{N-1}	Probability simplex in $\mathbb{R}^N$ (§1).
H_0	Zero-sum hyperplane $\{x \in \mathbb{R}^N : \sum_i x_i = 0\}$ — the natural tangent space (§2).
S_+^{N-1}	Positive orthant of the unit sphere (§3).
u	Uniform distribution $u = \frac{1}{N}(1, \dots, 1)$, the centroid of Δ^{N-1}.
$\psi,\ \varphi$	Square-root and squaring maps; mutual inverses between Δ^{N-1} and S_+^{N-1} (§3).
$g_{FR},\ g_S$	Fisher–Rao metric on Δ^{N-1}; round metric on S_+^{N-1} (§4).
$\langle \cdot, \cdot \rangle$	Ambient Euclidean inner product on $\mathbb{R}^N$.
ν	(for $N = 4$ only) Unit normal to H_0 in $\mathbb{R}^4$: $\nu = \frac{1}{2}(1, 1, 1, 1)$ (§6).
$\varepsilon_{ijk\ell}$	Levi-Civita symbol on $\mathbb{R}^4$; $\varepsilon_{1234} = +1$ (§6).

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1. The probability simplex and its gauge redundancy

A discrete probability distribution over N outcomes is a vector

$$p = (p_1, p_2, \dots, p_N) \in \mathbb{R}^N$$

with $p_i \geq 0$ and $\sum_{i=1}^N p_i = 1$. The set of such vectors is the **probability simplex**

$$\Delta^{N-1} = \{p \in \mathbb{R}^N : p_i \geq 0,\ \textstyle\sum_i p_i = 1\},$$

an $(N-1)$-dimensional convex polytope with N vertices at the standard basis vectors $e_1, \ldots, e_N$ (the deterministic distributions).

The dimensional mismatch is the first thing to notice: the simplex is $(N-1)$-dimensional but its natural description uses N coordinates. One coordinate is redundant — fixed by the others through $p_N = 1 - p_1 - \cdots - p_{N-1}$.

The standard move is to **eliminate** the redundant coordinate. Work in $(p_1, \ldots, p_{N-1})$, treat p_N as derived. This gives an unconstrained chart on the interior of the simplex, at the cost of singling out the N-th outcome. The simplex's natural permutation symmetry is broken; every subsequent operation either carries the symmetry around manually or repairs it after the fact.

We will refuse the elimination.

2. The zero-sum hyperplane

Define

$$H_0 = \{x \in \mathbb{R}^N : \textstyle\sum_i x_i = 0\}.$$

This is the **zero-sum hyperplane**: a vector subspace of $\mathbb{R}^N$ of dimension $N-1$.

Two facts make H_0 the natural object.

Tangent space. Every tangent vector to Δ^{N-1} at any interior point lies in H_0. If $p(t)$ is a smooth curve in Δ^{N-1}, differentiating $\sum_i p_i(t) = 1$ gives $\sum_i \dot{p}_i = 0$.

Differences of distributions. If $p, q \in \Delta^{N-1}$ then $p - q \in H_0$. Score functions, parameter perturbations, gradients of functions on the simplex — all live in H_0.

H_0 is therefore the natural arena for linear algebra on the simplex. It is a genuine Euclidean vector space, closed under addition, scaling, inner product, and rotation. No Lagrange multipliers; no constraint surface; no special handling of boundary versus interior.

The **uniform distribution**

$$u = \tfrac{1}{N}(1, 1, \ldots, 1)$$

is the centroid of Δ^{N-1} and the unique simplex point fixed by all coordinate permutations. The displacement map

$$\Delta^{N-1} \longrightarrow H_0, \qquad p \longmapsto p - u$$

is the **centered simplicial chart**. It is overcomplete (still N coordinates), but the coordinates satisfy a single linear relation and the chart preserves the full permutation symmetry by construction.

3. The square-root map and the amplitude sphere

Define the **square-root map**

$$\psi : \Delta^{N-1} \longrightarrow S_+^{N-1}, \qquad \psi(p) = (\sqrt{p_1}, \sqrt{p_2}, \dots, \sqrt{p_N})$$

where

$$S_+^{N-1} = \{a \in \mathbb{R}^N : a_i \geq 0,\ \textstyle\sum_i a_i^2 = 1\}$$

is the **positive orthant of the unit sphere** in $\mathbb{R}^N$. The components $a_i = \sqrt{p_i}$ are called **amplitudes**.

The inverse is coordinate-wise squaring:

$$\varphi : S_+^{N-1} \longrightarrow \Delta^{N-1}, \qquad \varphi(a) = (a_1^2, a_2^2, \dots, a_N^2).$$

Constraints transfer cleanly: $a_i \geq 0$ since $p_i \geq 0$; and $\sum a_i^2 = \sum p_i = 1$.

Why this is more than renaming. The simplex normalization $\sum p_i = 1$ is *linear* in p — it cuts out an affine hyperplane. The sphere normalization $\sum a_i^2 = 1$ is *quadratic* in a — it cuts out a sphere. Affine hyperplanes carry no rotational symmetry beyond what the ambient space gives them. Spheres are homogeneous under their full rotation group $\mathrm{SO}(N)$. Trading a linear constraint for a quadratic one is trading flat geometry for curved — and curved is what statistics wants, because probability distributions are not equidistant in any meaningful sense of distinguishability.

Boundary caveat. ψ is continuous on all of Δ^{N-1} but smooth only on the open interior, where every $p_i > 0$. At boundary points (some $p_i = 0$) the derivative $\partial\psi_i/\partial p_i = 1/(2\sqrt{p_i})$ diverges, and the simplex itself acquires corner structure where the tangent space is not H_0. All subsequent statements in this paper are made on the open interior, where the chart is smooth.

4. Fisher–Rao as the round metric, pulled back

The unit sphere carries the **round metric** — the restriction of the ambient Euclidean inner product:

$$g_S = \sum_{i=1}^{N} (da_i)^2 \Big|_{S^{N-1}}.$$

Pull this back through ψ. With $a_i = \sqrt{p_i}$:

$$da_i = \frac{dp_i}{2\sqrt{p_i}}, \qquad (da_i)^2 = \frac{(dp_i)^2}{4p_i}.$$

Summing:

$$\psi^* g_S = \sum_i (da_i)^2 = \frac{1}{4} \sum_i \frac{(dp_i)^2}{p_i} = \frac{1}{4}\, g_{FR}(p),$$

where

$$g_{FR}(p) = \sum_i \frac{(dp_i)^2}{p_i}$$

is the **Fisher–Rao metric** [Fisher, 1925; Rao, 1945] on the simplex — the standard convention used throughout this paper. The metric is unique up to scale under invariance by sufficient statistics [Chentsov, 1982]; see [Amari & Nagaoka, 2000] for the canonical modern exposition. Some authors absorb the factor $1/4$ into g_{FR} to make $\psi^* g_S = g_{FR}$ exactly; we keep it explicit because it simplifies the bookkeeping in §5.

Three immediate consequences:

Distances are arc lengths. The Fisher–Rao distance between $p, q \in \Delta^{N-1}$ is

$$d_{FR}(p,q) = 2 \arccos\left(\sum_i \sqrt{p_i\, q_i}\right) = 2\angle(\psi(p), \psi(q)),$$

twice the angle between the amplitude vectors on the sphere.

The Bhattacharyya coefficient is the amplitude inner product. The quantity $BC(p,q) = \sum_i \sqrt{p_i q_i}$ from classical statistics [Bhattacharyya, 1943] is exactly $\langle \psi(p), \psi(q) \rangle$, the Euclidean inner product in $\mathbb{R}^N$.

Geodesics are great circles, pulled back. The Fisher–Rao geodesic from p to q is the φ-image of the great-circle arc from $\psi(p)$ to $\psi(q)$ on the sphere.

5. The tangent space at the uniform

The uniform distribution has amplitude vector

$$\psi(u) = \tfrac{1}{\sqrt{N}}(1, 1, \ldots, 1).$$

The tangent space to S^{N-1} at $\psi(u)$ consists of vectors in $\mathbb{R}^N$ orthogonal to $\psi(u)$ — that is, vectors with $\sum_i v_i = 0$. So

$$T_{\psi(u)} S^{N-1} = H_0,$$

and via ψ_*^{-1}, the Fisher–Rao tangent space at u is also H_0. This matches the **centered chart of §2** directly: linearizing the affine constraint $\sum p_i = 1$ at u gives the linear constraint $\sum x_i = 0$, which is the defining equation of H_0. The two charts of §§2–3 therefore converge at the uniform, both identifying $T_u \Delta^{N-1}$ with H_0 as a vector space — differing only by the scalar Jacobian $\psi_*\big|_u = (\sqrt{N}/2) \cdot \mathrm{id}_{H_0}$ obtained from differentiating ψ at u.

Evaluating g_{FR} at u:

$$g_{FR}(v,w)\big|_u = \sum_i \frac{v_i\, w_i}{u_i} = N \sum_i v_i w_i = N \langle v, w \rangle_{\mathbb{R}^N}, \qquad v, w \in H_0.$$

Fisher–Rao at the uniform is therefore N times the ambient Euclidean inner product, restricted to H_0. (Equivalently, by §4, $\psi^* g_S\big|_u = (N/4)\langle\cdot,\cdot\rangle$ — the factor of $1/4$ from §4 and the factor of N from evaluating $1/u_i$ multiply to give $N/4$.)

Tangent (isotropy) isometry group at the uniform. Rotations of S^{N-1} that fix $\psi(u)$ form the subgroup

$$\mathrm{Stab}(\psi(u)) = \mathrm{SO}(H_0) \cong \mathrm{SO}(N-1)$$

acting linearly on the tangent space. *Proof of the linear-isometry claim.* The round metric g_S is $O(N)$-invariant; in particular, the stabilizer of any point $a \in S^{N-1}$ acts on $T_a S^{N-1}$ by linear isometries of g_S. Since $\psi^* g_S = g_{FR}/4$ (§4), the same linear maps preserve the bilinear form $g_{FR}|_u$ on $T_u\Delta^{N-1} = H_0$. Therefore $\mathrm{SO}(H_0)$ is the **full linear (isotropy) isometry group of the Fisher–Rao tangent inner product at the uniform** — the group of linear self-maps of H_0 preserving $g_{FR}|_u$, equivalently the isotropy representation on $T_{\psi(u)}S^{N-1}$. The discrete subgroup generated by coordinate permutations is the symmetric group S_N, acting on H_0 as the standard $(N-1)$-dimensional irreducible representation.

Scope of the claim. This is the **tangent/isotropy** statement — symmetries of the bilinear form on $T_u\Delta^{N-1}$. The **global** isometry group of (Δ^{N-1}, g_{FR}) as a Riemannian manifold is strictly smaller: a generic element of $\mathrm{SO}(H_0)$ does not lift through ψ to a sphere rotation preserving the positive orthant S^{N-1}_+, so it does not extend to a global Fisher–Rao isometry of the simplex. The automatic global symmetry is only the relabeling action of S_N. Throughout this paper and the trilogy, "the isometry group at u" — and the corresponding $\mathrm{SO}(H_0)$ label — refers to the linear isometry group of $g_{FR}|_u$ on the tangent space, unless stated otherwise.

This is the key fact the chart was built to expose:

> **The linear isometry group of the Fisher–Rao tangent inner product at the uniform is $\mathrm{SO}(H_0)$ — the full rotation group of the zero-sum hyperplane.**

The simplicial rotation operator and the Fisher–Rao tangent isometry at u are not analogues; they are the same Lie group acting on the same tangent space.

6. Intrinsic vector algebra on H_0

Because H_0 is a Euclidean vector space, ordinary finite-dimensional linear algebra applies on it without modification. Three operations are worth naming.

Inner product. For $v, w \in H_0$,

$$\langle v, w\rangle = \sum_{i=1}^{N} v_i w_i.$$

Restricted to H_0 this is positive-definite. It is $1/N$ of the Fisher–Rao inner product at the uniform (by §5).

Cross product (the tetrahedral case $N = 4$). When $N = 4$, H_0 is three-dimensional. Fix the unit normal to H_0 in $\mathbb{R}^4$:

$$\nu = \tfrac{1}{2}(1,1,1,1) \in \mathbb{R}^4, \qquad |\nu| = 1, \qquad \nu \perp H_0.$$

Define the **simplicial cross product** $\times_S : H_0 \times H_0 \to H_0$ by

$$(v \times_S w)_i = \sum_{j,k,\ell=1}^{4} \varepsilon_{ijk\ell}\, v_j\, w_k\, \nu_\ell, \qquad \varepsilon_{1234} = +1,$$

where ε is the standard Levi-Civita symbol on $\mathbb{R}^4$. Equivalently, $v \times_S w$ is the Hodge dual of $v \wedge w \wedge \nu$ in $\mathbb{R}^4$.

The product is bilinear and antisymmetric in (v, w) from the antisymmetry of ε in (j, k). It lands in H_0: the inner product

$$\langle v \times_S w, \nu \rangle = \sum_{i,j,k,\ell} \varepsilon_{ijk\ell}\, \nu_i\, v_j\, w_k\, \nu_\ell$$

vanishes because ε is antisymmetric in (i, ℓ) while $\nu_i \nu_\ell$ is symmetric. In any orthonormal frame on H_0 (such as the Helmert frame of §7.3, verified there by direct computation), $\times_S$ agrees with the ordinary three-dimensional Euclidean cross product. The tetrahedral case is not an analogy to 3D vector algebra; the two algebras are isomorphic.

Rodrigues rotation. For a unit vector $n \in H_0$ and angle θ, the operator

$$R_{n,\theta}(v) = (\cos\theta)\, v + (\sin\theta)\, n \times_S v + (1 - \cos\theta)\, \langle n, v \rangle\, n$$

is a rotation of H_0 by angle θ about the axis n. For $N = 4$ this is the Fisher–Rao isometry of Δ^3 at the uniform, rotating tangent directions by θ.

Higher N does not admit a binary cross product (Hodge duality on H_0 no longer lands in H_0), but the generators of $\mathrm{SO}(H_0)$ are still the elements of $\Lambda^2 H_0$ and Rodrigues generalizes via the matrix exponential. The case $N = 4$ is special only because $\dim H_0 = 3$.

7. Worked examples

7.1. $N = 2$: Bernoulli (the 1-simplex)

Δ^1 is the segment from $e_1 = (1, 0)$ to $e_2 = (0, 1)$, with H_0 the line spanned by $(1, -1)/\sqrt{2}$.

The square-root map sends Δ^1 to the quarter-circle S^1_+. Parameterize by angle $\theta \in [0, \pi/2]$:

$$\psi(p) = (\cos\theta, \sin\theta), \qquad p = (\cos^2\theta, \sin^2\theta).$$

θ	p_1	p_2	a_1	a_2
0	1	0	1	0
$\pi/6$	$3/4$	$1/4$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$1/2$	$1/2$	$1/\sqrt{2}$	$1/\sqrt{2}$

θ	p_1	p_2	a_1	a_2
$\pi/3$	$1/4$	$3/4$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1	0	1

The Fisher–Rao distance between θ_1 and θ_2 is $2|\theta_1 - \theta_2|$. The total Fisher–Rao length of Δ^1 is π. Notice that the gap in p_1 between $\theta = \pi/4$ and $\theta = \pi/3$ (fair to mildly biased) is 0.25 — the same as the gap between $\theta = \pi/3$ and $\theta = \pi/2$ (mildly biased to deterministic). But the Fisher–Rao distance is $\pi/6$ for the first pair and $\pi/3$ for the second: the second is twice as far. The angular coordinate tracks **statistical distinguishability**; the probability coordinate does not.

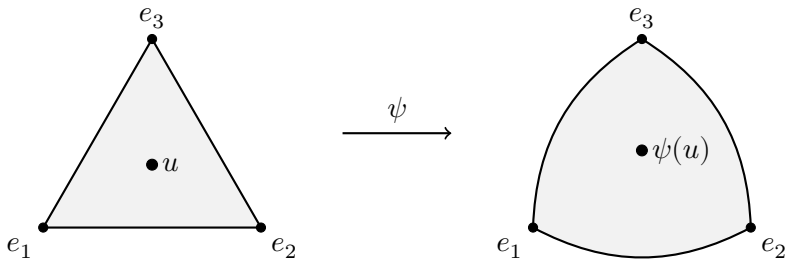
7.2. $N = 3$: categorical over three outcomes (the 2-simplex)

Δ^2 is an equilateral triangle in $\mathbb{R}^3$ with vertices e_1, e_2, e_3 and centroid $u = (1/3, 1/3, 1/3)$. H_0 is the plane orthogonal to $(1, 1, 1)$.

ψ sends Δ^2 to the positive octant of S^2 — a curved spherical triangle. The vertices coincide (they were already unit vectors). The centroid lifts:

$$u = (1/3, 1/3, 1/3),\ |u| = 1/\sqrt{3}; \qquad \psi(u) = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}),\ |\psi(u)| = 1.$$

ψ "inflates" the flat triangle's interior radially onto the spherical surface. Edges (where some $p_i = 0$) map to great-circle arcs (where the same $a_i = 0$).



(a) $\Delta^2 \subset \mathbb{R}^3$ (b) $\psi(\Delta^2) = S^2_+$

Figure 1: The square-root map $\psi : \Delta^2 \to S^2_+$ for $N = 3$. (a) The probability simplex Δ^2, an equilateral triangle in $\mathbb{R}^3$ with vertices at the standard basis vectors e_1, e_2, e_3 and centroid $u = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ at radial distance $1/\sqrt{3}$ from the origin. (b) The spherical triangle $\psi(\Delta^2) = S^2_+$, the positive octant of the unit sphere, sharing the same vertices (already unit vectors) but with the centroid lifted radially to $\psi(u) = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$ on the sphere. The flat edges (where some $p_i = 0$) become the spherical triangle's great-circle arcs (where the corresponding amplitude $a_i = 0$).

(p_1, p_2, p_3)	(a_1, a_2, a_3)
$(1, 0, 0)$	$(1, 0, 0)$
$(1/2, 1/2, 0)$	$(1/\sqrt{2}, 1/\sqrt{2}, 0)$
$(1/2, 1/4, 1/4)$	$(1/\sqrt{2}, 1/2, 1/2)$
$(1/3, 1/3, 1/3)$	$(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$

(p_1, p_2, p_3)	(a_1, a_2, a_3)
$(0.8, 0.1, 0.1)$	$(0.894, 0.316, 0.316)$

Fisher–Rao distances are arc lengths between amplitude images. For instance,

$$d_{FR}(u, e_1) = 2\arccos(\langle\psi(u), e_1\rangle) = 2\arccos(1/\sqrt{3}) \approx 1.911.$$

The cyclic group $\mathbb{Z}/3 \subset S_3$ acts on H_0 as rotation by $2\pi/3$ about the axis through $\psi(u)$.

7.3. $N = 4$: tetrahedral (the 3-simplex)

Δ^3 has four vertices and centroid $u = (1/4, 1/4, 1/4, 1/4)$. H_0 is three-dimensional. $\psi(\Delta^3)$ is a curved spherical tetrahedron — the positive orthant of S^3.

This is the case in which "the simplicial frame is just Euclidean 3-space" is literally true. Choose the **Helmert frame** [Helmert, 1876] on H_0:

$$e'_1 = \tfrac{1}{\sqrt{2}}(1, -1, 0, 0), \quad e'_2 = \tfrac{1}{\sqrt{6}}(1, 1, -2, 0), \quad e'_3 = \tfrac{1}{\sqrt{12}}(1, 1, 1, -3).$$

These vectors are mutually orthogonal, of unit length, all in H_0. The map

$$H_0 \xrightarrow{\sim} \mathbb{R}^3, \qquad v'_1 e'_1 + v'_2 e'_2 + v'_3 e'_3 \longmapsto (v'_1, v'_2, v'_3)$$

is an isometric isomorphism. Under it the simplicial cross product $\times_S$ on H_0 becomes the ordinary $\mathbb{R}^3$ cross product, and the simplicial Rodrigues operator becomes the ordinary 3D Rodrigues formula.

Verification. Compute $e'_1 \times_S e'_2$ from the formula of §6 with $\nu = (1, 1, 1, 1)/2$ and $\varepsilon_{1234} = +1$. Expanding the sum over $\{j, k, \ell\} = \{2, 3, 4\}$ as a cofactor identity:

$$(e'_1 \times_S e'_2)_1 = (e'_1)_2[(e'_2)_3\,\nu_4 - (e'_2)_4\,\nu_3] - (e'_1)_3[\cdots] + (e'_1)_4[\cdots].$$

Since $(e'_1)_3 = (e'_1)_4 = 0$, only the first bracket contributes:

$$(e'_1 \times_S e'_2)_1 = (-\tfrac{1}{\sqrt{2}})[(-\tfrac{2}{\sqrt{6}})(\tfrac{1}{2}) - 0] = (-\tfrac{1}{\sqrt{2}})(-\tfrac{1}{\sqrt{6}}) = \tfrac{1}{\sqrt{12}} = (e'_3)_1.$$

The remaining three components, computed analogously, also match e'_3 — so $e'_1 \times_S e'_2 = e'_3$. Antisymmetry and analogous computations on the other two pairs give the full cyclic identities

$$e'_1 \times_S e'_2 = e'_3, \qquad e'_2 \times_S e'_3 = e'_1, \qquad e'_3 \times_S e'_1 = e'_2.$$

This is the standard right-hand rule of 3D Euclidean cross product, holding verbatim in the Helmert frame on $H_0 \subset \mathbb{R}^4$.

The orientation-preserving tetrahedral group A_4 acts on H_0 as a discrete subgroup of SO(3) of order 12 (rotational symmetries of the regular tetrahedron). Including reflections gives $S_4 \cong T_d$ of

order 24. The cyclic subgroup $\mathbb{Z}/3 \subset A_4$ generated by rotation about a vertex-centroid axis is the rotational symmetry visible in §7.2's three-outcome case, lifted to the full tetrahedral frame.

This is the case used in the Quadray / synergetics tradition [Fuller, 1975], and it is the case in which all of ל1's claims can be checked by hand against ordinary 3D vector calculus.

8. The chart, as a working toolkit

ל1's toolkit, compressed:

Object	Standard chart	Native simplicial chart
Probability vector	$(p_1, \dots, p_{N-1})$; p_N derived	$(p_1, \dots, p_N)$ with $\sum p_i = 1$ as gauge
Tangent space	Constraint-aware, varies with chart	H_0 (fixed Euclidean $(N-1)$-space)
Metric	g_{FR} on a constrained manifold	$4 \cdot$ (round sphere metric), pulled back via ψ
Tangent isometry group at u	Computed via Chentsov uniqueness	$\mathrm{SO}(H_0)$, by direct identification (linear isometries of $g_{FR}\|_u$ on H_0; the global isometry group of (Δ^{N-1}, g_{FR}) is strictly smaller)
Tangent inner product at u	$\sum(v_i w_i / p_i)$ at p	$N\langle v, w\rangle_{\mathbb{R}^N}$ on H_0 (ambient Euclidean, scaled by N)
Bhattacharyya coefficient	Defined separately	$\langle \psi(p), \psi(q)\rangle$
Geodesics	Solved via Fisher–Rao ODE	φ-image of great circles

The native chart removes machinery rather than adding it. Two maps — the centering $p \mapsto p - u$ and the square root ψ — expose the simplex's Euclidean tangent space, sphere metric, and rotation group simultaneously. The chart, the map ψ, the zero-sum hyperplane H_0, the identification of Fisher–Rao with the pulled-back round metric, and the intrinsic vector algebra on H_0 are complete in themselves as a coordinatization of the categorical probability simplex.

ל2 develops the cumulant tower of the categorical log-partition on this chart; ל3 reads the chart through Pontryagin character orthogonality. Neither modifies ל1's toolkit; both inherit it.

9. References

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